

Global Stock Reallocation — Decision Memo

Specialty chemicals distribution network · 30 warehouses across India · real city locations & freight/warehousing benchmarks

Problem

A specialty-chemicals company's Indian warehouse network shows uneven inventory distribution: several warehouses sit on surplus stock (tying up capital, some approaching expiry) while others are understocked relative to local demand, risking stockouts. There was no systematic process to identify these imbalances and rebalance the network.

Method

Built a linear programming (LP) model in Python (PuLP) over 30 warehouses spanning real Indian cities (including genuine specialty-chemicals hubs like Vadodara, Ankleshwar, and Vapi). Inter-warehouse transfer costs are grounded in real distances (computed from real coordinates) and real published Indian freight benchmarks (~Rs.20-30/km for a medium FTL truck); holding costs are grounded in real Indian warehousing rent benchmarks (Rs.100-300/pallet/month, varying by city tier). The model minimizes total network cost — transfer cost + holding cost on unallocated surplus + a soft write-off penalty on unmoved near-expiry stock + a stockout-risk cost on unmet deficit — and was benchmarked against a do-nothing baseline and a naive rule-based heuristic, then stress-tested against $\pm 15\%$ demand forecast error.

1,936 Units reallocated	16.5% Idle stock reduction	8 of 8 Warehouses de-risked	Rs.7.14L Net Rs. saved (6-mo)
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Key Results

- Reallocated 1,936 units — entirely sourced from near-expiry surplus warehouses — fully closing the shortfall at all 8 at-risk (stockout-facing) warehouses across the network.
- Cut idle surplus stock by 16.5% (from 11,754 to 9,816 units): the model correctly prioritized moving near-expiry stock (highest write-off risk) ahead of healthy surplus, which is why the reduction is concentrated rather than spread evenly.
- Delivered a net saving of ₹7,13,538 over a 6-month horizon (holding cost saved + stockout cost avoided + expiry write-off avoided, net of transfer cost paid), against a do-nothing baseline.
- The LP plan outperformed a naive sort-and-match heuristic by ₹41,497 (6.2% more savings captured) on identical data, by routing shipments through cheaper real freight lanes rather than the nearest-priority match.
- Plan held up under demand stress-testing: net savings ranged ₹5.11L–₹8.39L across a $\pm 15\%$ demand swing, staying solidly positive and fully de-risking all 8 at-risk warehouses in every scenario.

Recommendation

Prioritize moving near-expiry surplus first — the model treats this as a soft but heavily weighted priority (a hard quota was tested and found infeasible at this network size, since near-expiry surplus can exceed what deficit warehouses can actually absorb). Treat all other surplus-to-deficit transfers as a cost-benefit decision, not a blanket policy — the model identifies exactly which lanes pay for themselves via real freight-cost data, which is the core value it adds over an intuition-based reallocation process.

Scope & Limitations

Warehouse locations, inter-city distances, and freight/warehousing cost benchmarks are real and sourced from published Indian logistics-industry data. Current stock levels, exact demand, and expiry dates are synthetic — no company publishes this data — though demand is scaled using each city's real Metro/Tier-2 classification rather than picked arbitrarily. This is a self-built proof-of-concept methodology, not a real company engagement; in a production setting, thresholds (stockout/waste penalties, holding-cost horizon) would be calibrated against a specific company's actual historical stockout and spoilage costs.